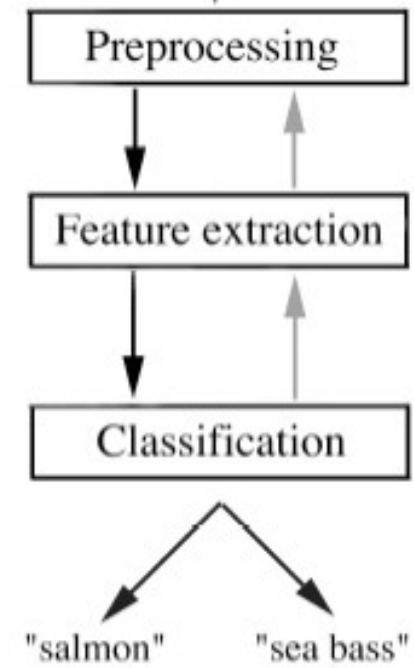


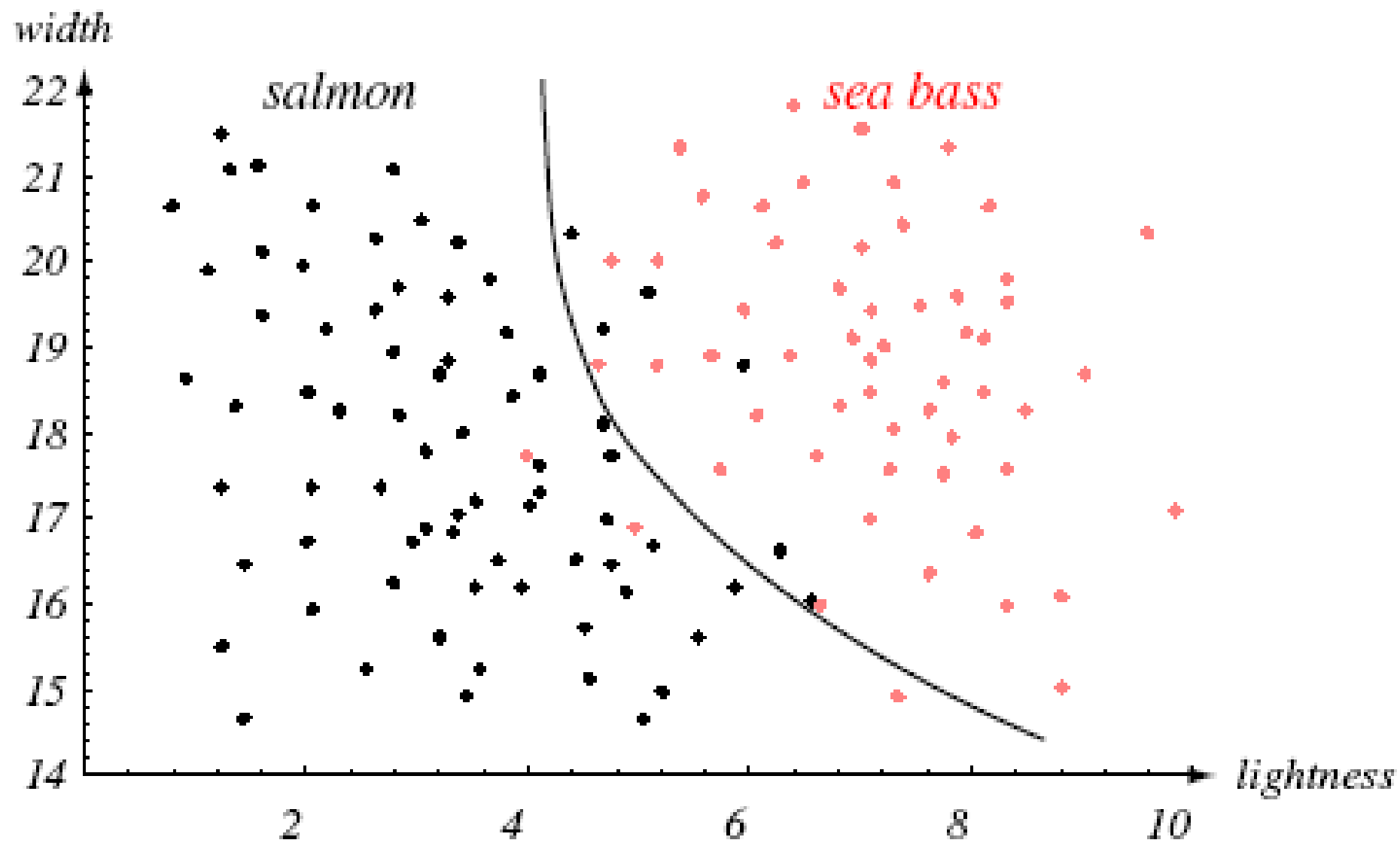
Automatic Image Analysis

Bayesian Probability Estimation

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Probability Theory

Let Ω be the sample space,
i.e. the set Ω of all possible discrete outcomes ω .

A function $P: P(\Omega) \rightarrow \mathbb{R}$ is called a **probability** if and only if the following statements are valid:

1. $P(\omega) \geq 0$
2. $\sum_{\omega \in \Omega} P(\omega) = 1$

Note, that from 1. and 2. directly follows that $P(\omega) \leq 1$.

In this case, the probability of an event $E \subseteq P(\Omega)$ is given by

$$P(E) = \sum_{\omega \in E} P(\omega)$$

Probability Theory

If Ω is a continuous space, a cumulative distribution function $F: \Omega \rightarrow \mathbb{R}$ exists, with $F(x) = P(X \leq x)$ that is monotonically increasing where

$$\lim_{x \rightarrow -\infty} F(x) = 0$$

$$\lim_{x \rightarrow +\infty} F(x) = 1$$

The function F exists for all random variables. If Ω is continuous, then

$p(x) = \frac{\partial F(x)}{\partial x}$ is called the probability density function. Note, that

$$1. \quad \forall x \in \Omega: p(x) \geq 0$$

$$2. \quad \int_{\Omega} p(x) dx = 1$$

$$3. \quad P(x \in E) = \int_{x \in E} p(x) dx$$

From 1. & 2. does NOT follow that

$$\forall x \in \Omega: p(x) \leq 1$$

but

$$\forall E \subseteq \Omega: P(E) \leq 1$$

Gaussian Distribution

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

- Important probability density function

- Related to many real-world phenomena (e.g. noise)

- Central limit theorem

- Nice analytical properties

- Closed-form solution for parameter estimates

$$\mu = E[x] = \int x p(x) dx \quad \sigma^2 = E[(x - \mu)^2] = \int (x - \mu)^2 p(x) dx$$

- Analytical derivatives exist

$$\frac{\partial p(x)}{\partial x} = -\left(\frac{x - \mu}{\sigma^2}\right) \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2}\left(\frac{x - \mu}{\sigma}\right)^2\right) = -\left(\frac{x - \mu}{\sigma^2}\right) p(x)$$

- Largest entropy among distributions of same mean and variance

$$H(x) = -\int x \ln(p(x)) dx$$

Probability Theory

Joint probability

$$P(A=a, B=b)$$

Conditional probability

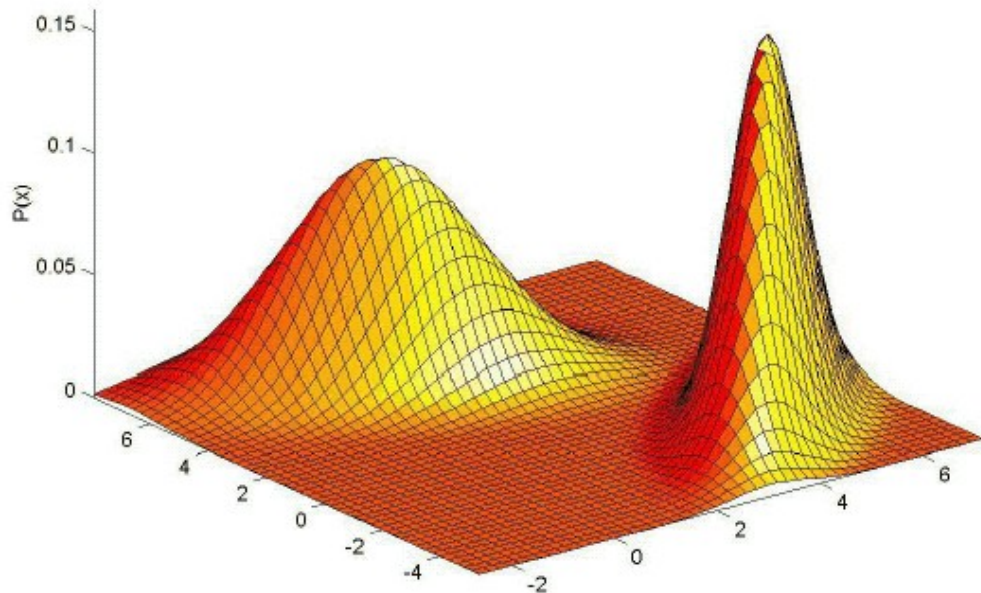
$$P(A=a|B=b)$$

Probability that

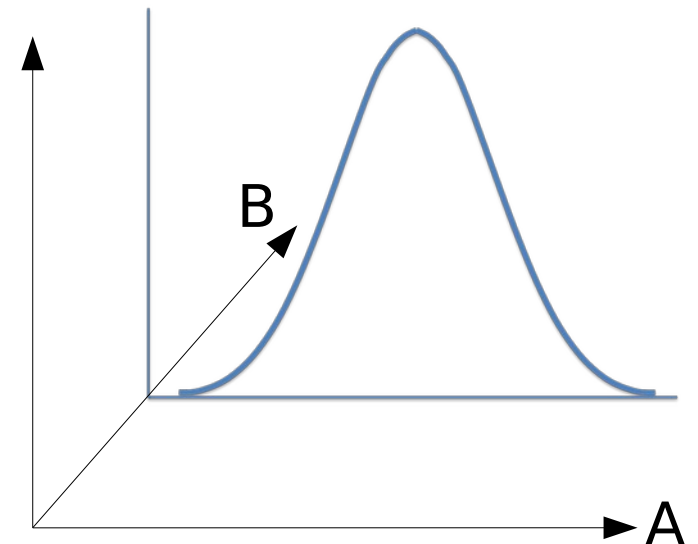
event A and B occur together.

event A occurs given event B.

→ Two dimensional



→ One dimensional



Probability Theory

Marginalization

$$P(A=a) = \sum_b P(A=a, B=b) \quad p(x) = \int p(x, y) dy$$

Conditioning

$$P(A, B) = P(A|B) P(B)$$

$$p(x, y) = p(x|y) p(y)$$

$$P(A, B) = P(B|A) P(A)$$

$$p(x, y) = p(y|x) p(x)$$

Independence

$$P(A|B) = P(A)$$

$$p(x|y) = p(x)$$

$$P(A, B) = P(A) P(B)$$

$$p(x, y) = p(x) p(y)$$

Probability Theory

Bayes' Theorem

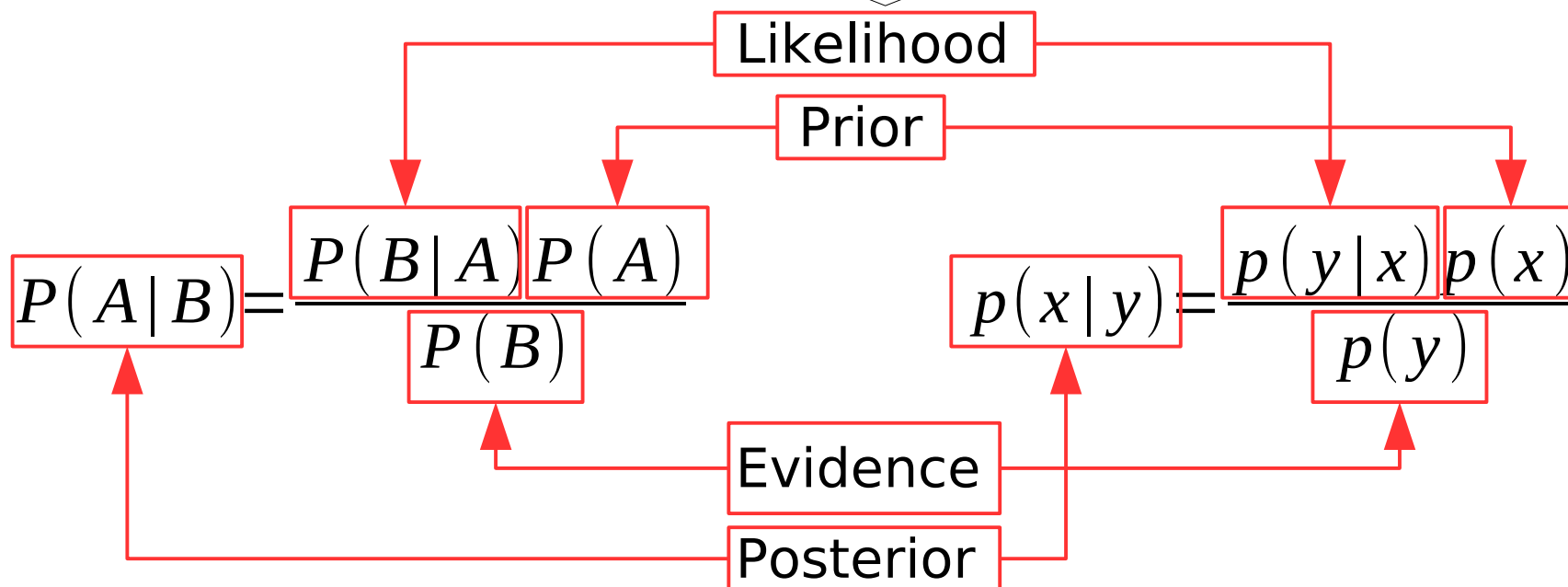
$$P(A, B) = P(A|B)P(B)$$

$$p(x, y) = p(x|y)p(y)$$

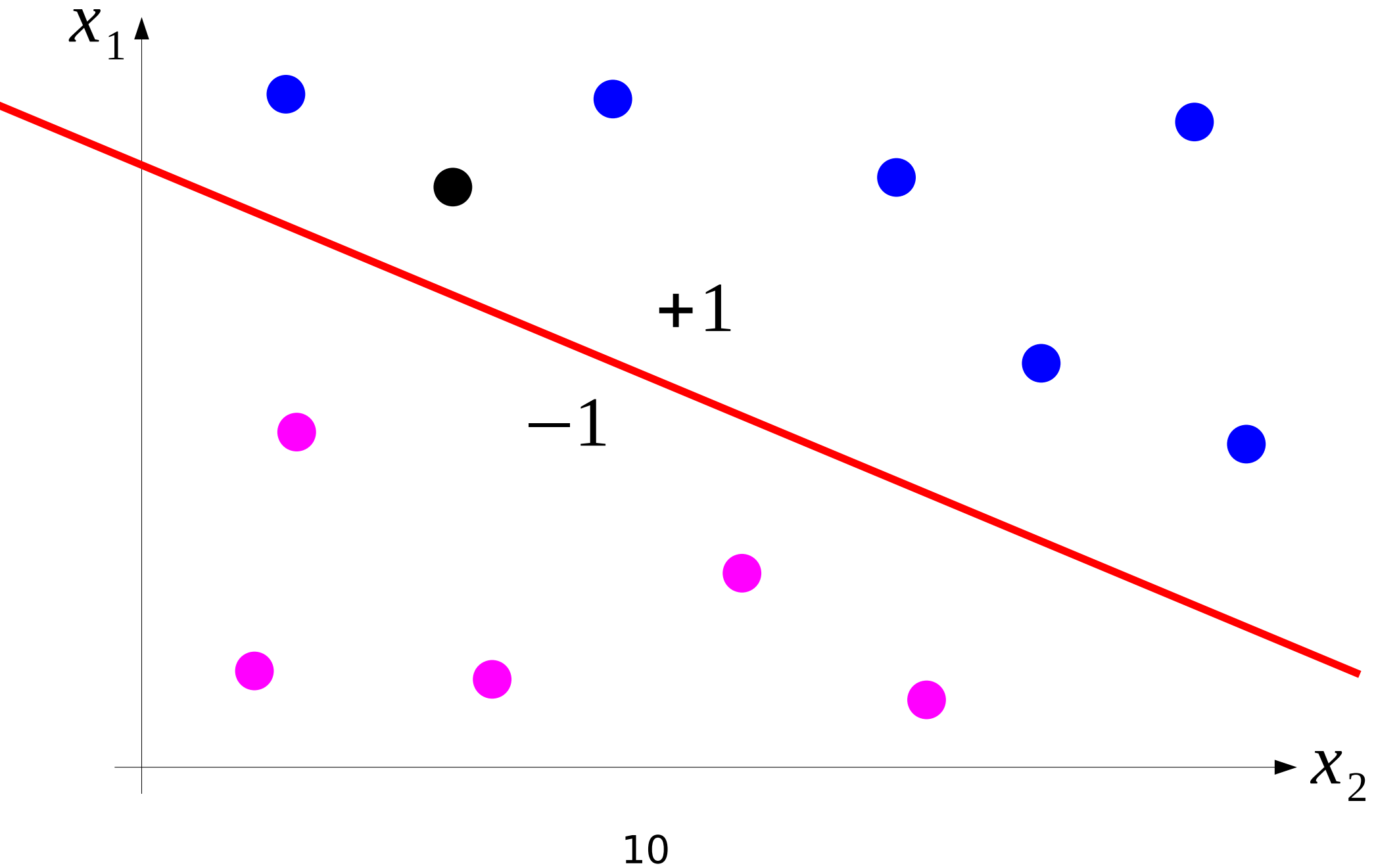
$$P(A, B) = P(B|A)P(A)$$

$$p(x, y) = p(y|x)p(x)$$

$$P(A|B)P(B) = P(B|A)P(A) \quad p(x|y)p(y) = p(y|x)p(x)$$



Typical Classifiers



Bayesian Decision Theory

Does this picture show a tiger?
Yes or No?

What if you know that this picture
was taken in Indian jungle?
... or Antarctica?



Let $\omega \in \Omega = \{\text{tiger}, \text{no tiger}\} = \{\omega_1, \omega_2\}$, then $P(\omega_1) + P(\omega_2) = 1$.

Decision rule based on a priori information (i.e. before observation):

Select ω_1 if $P(\omega_1) > P(\omega_2)$, otherwise select ω_2 .

- Prior probability
 - Not based on any (current) observations
 - Based on expert knowledge, given information, etc.
 - Often assumed to be uniform distributed

Bayesian Decision Theory

Does this picture show a tiger?
Yes or No?



Let $\mathbf{x}$ be observations, i.e. the picture itself or features contained in it.

Decision rule based on a posteriori information (i.e. after observation):

Select ω_1 if $P(\omega_1 | \mathbf{x}) > P(\omega_2 | \mathbf{x})$, otherwise select ω_2 .

- Posterior probability

- Based on current observations
- Often modelled via Bayes Theorem (see next slide)
- Discriminative approaches directly model the posterior.

Bayesian Decision Theory

Decision rule:

Select ω_1 if $P(\omega_1 | \mathbf{x}) > P(\omega_2 | \mathbf{x})$, otherwise select ω_2 .

More general:

Select ω^* , where $\omega^* = \underset{\omega}{\operatorname{argmax}} P(\omega | \mathbf{x}) = \underset{\omega}{\operatorname{argmax}} \frac{p(\mathbf{x} | \omega) P(\omega)}{p(\mathbf{x})}$
 $= \underset{\omega}{\operatorname{argmax}} p(\mathbf{x} | \omega) P(\omega)$

→ Maximum a posteriori (MAP) estimator

→ Naive Bayes: $p(\mathbf{x} | \omega) = \prod_{i=1}^F p(x_i | \omega)$ (independence assumptions)

If prior is uniform (i.e. all cases are equally likely):

Select ω^* , where $\omega^* = \underset{\omega}{\operatorname{argmax}} p(\mathbf{x} | \omega)$ → Maximum Likelihood (ML) estimator

Bayesian Decision Theory

Even more general:

Let $\lambda(\alpha_i | \omega_j)$ be the loss caused by action α_i given state ω_j .

New decision rule: Select action with minimal expected loss (risk).

Select α^* , where $\alpha^* = \underset{\alpha_i}{\operatorname{argmin}} R(\alpha_i | \mathbf{x}) = \underset{\alpha_i}{\operatorname{argmin}} \sum_{\omega} \lambda(\alpha_i | \omega) P(\omega | \mathbf{x})$

In general: **Discriminant functions**

Sample $\mathbf{x}$ is assigned to class / action / etc. ω_i if: $\forall i \neq j$:

$$g_i(\mathbf{x}) > g_j(\mathbf{x})$$

Discriminant Functions

MAP & Bayes Theorem:

$$g_i(\mathbf{x}) = P(\omega_i | \mathbf{x}) = \frac{p(\mathbf{x} | \omega_i) P(\omega_i)}{p(\mathbf{x})}$$

or

$$g_i(\mathbf{x}) = p(\mathbf{x} | \omega_i) P(\omega_i)$$

or

$$g_i(\mathbf{x}) = \ln(p(\mathbf{x} | \omega_i)) + \ln(P(\omega_i))$$

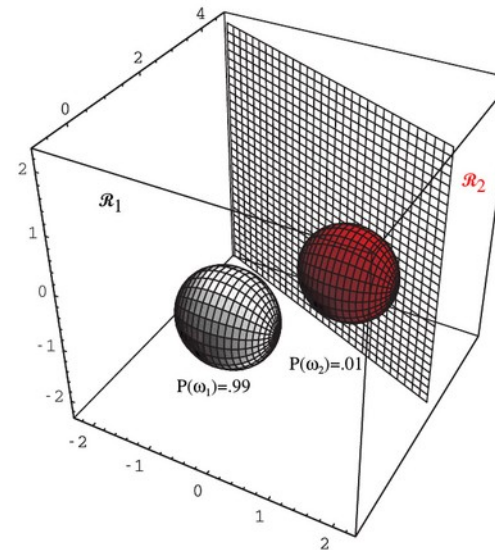
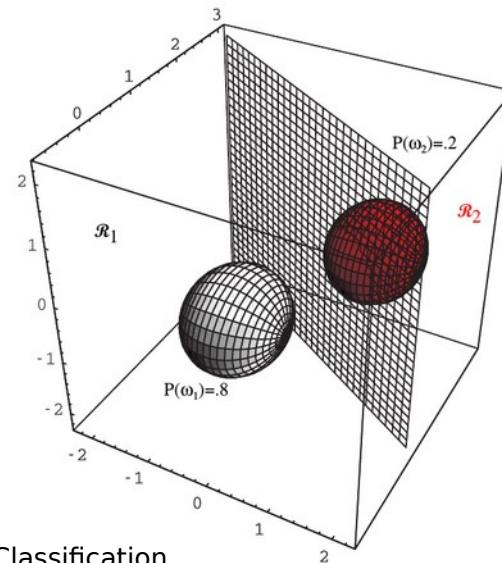
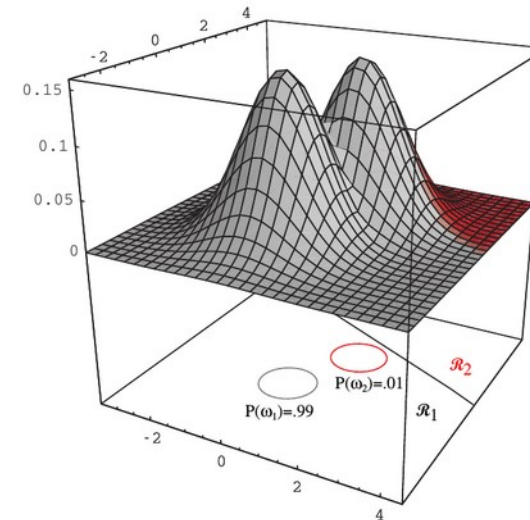
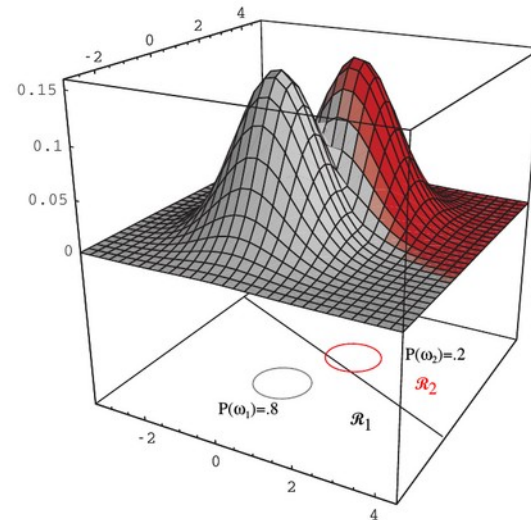
If likelihood modelled as Normal distribution:

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\boldsymbol{\Sigma}_i| + \ln(P(\omega_i))$$

Discriminant Functions

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\boldsymbol{\Sigma}_i| + \ln(P(\omega_i))$$

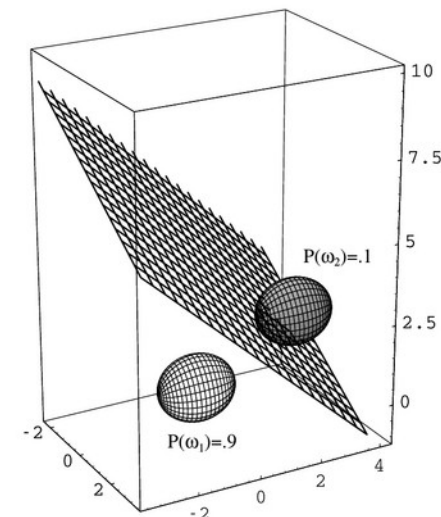
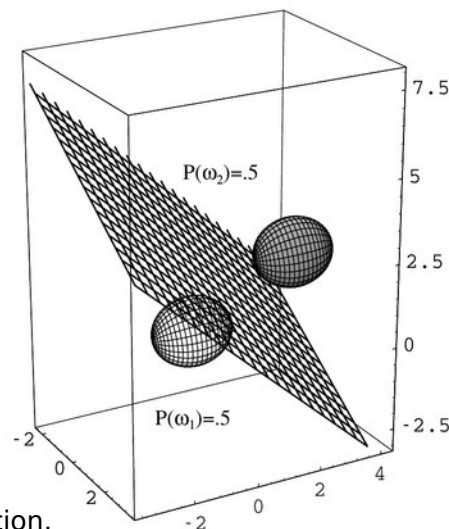
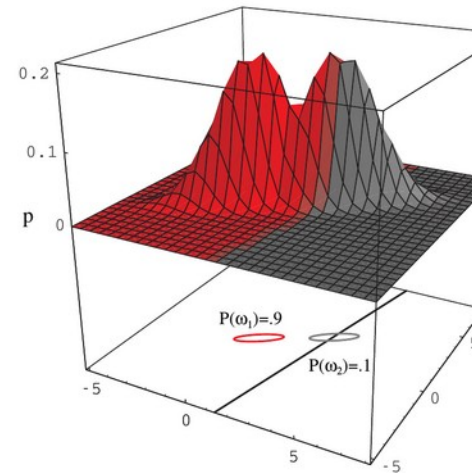
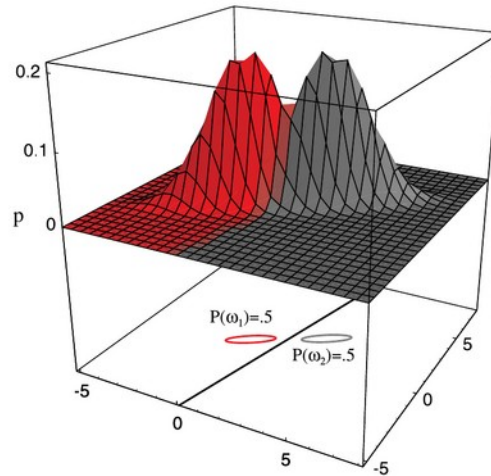
Case 1: $\boldsymbol{\Sigma}_i = \sigma^2 \mathbf{I}$



Discriminant Functions

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln|\boldsymbol{\Sigma}_i| + \ln(P(\omega_i))$$

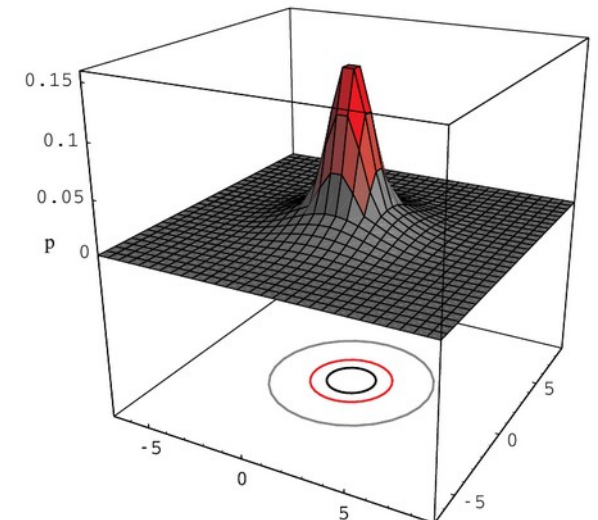
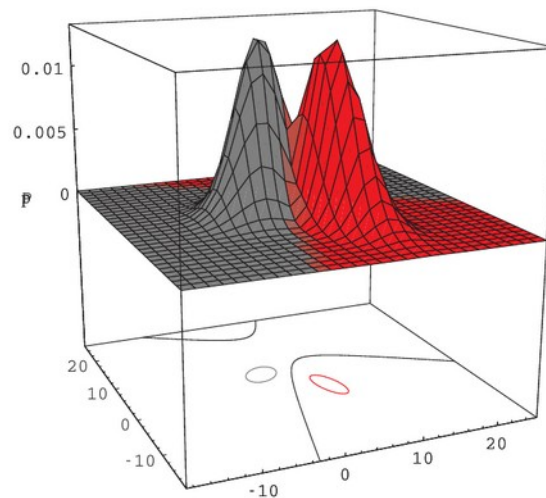
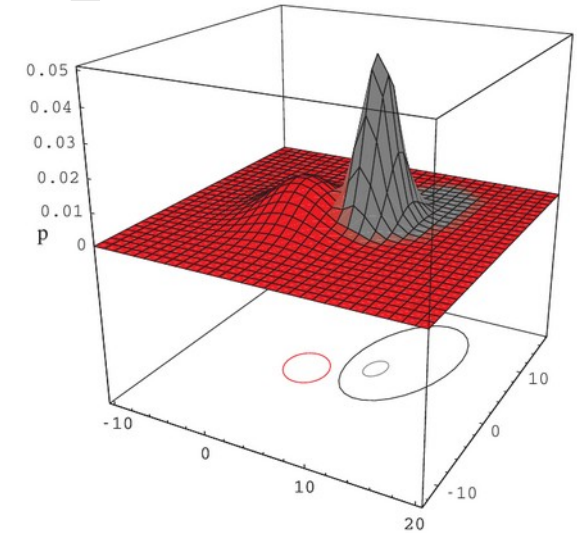
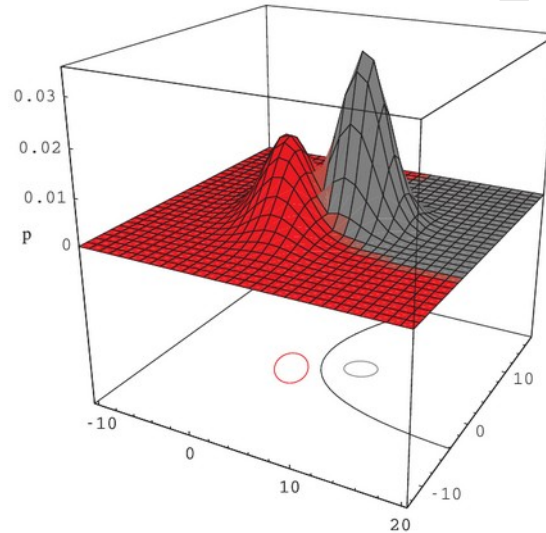
Case 2: $\boldsymbol{\Sigma}_i = \boldsymbol{\Sigma}$



Discriminant Functions

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\boldsymbol{\Sigma}_i| + \ln(P(\omega_i))$$

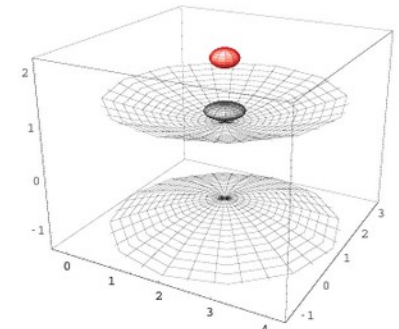
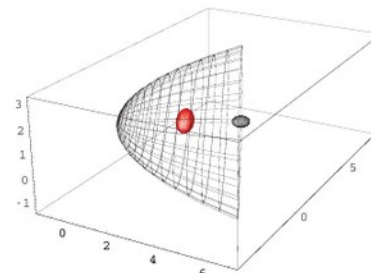
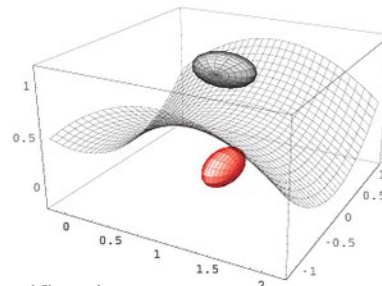
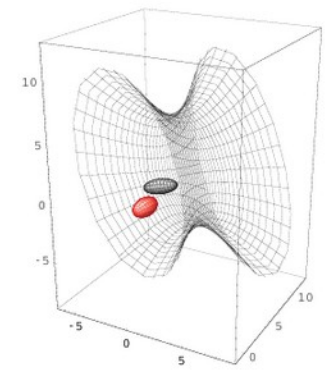
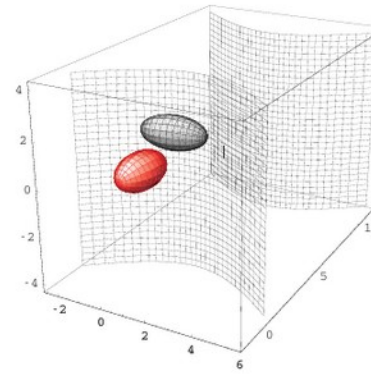
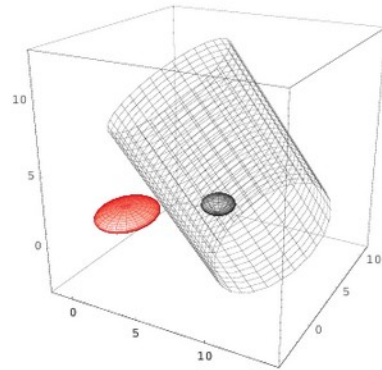
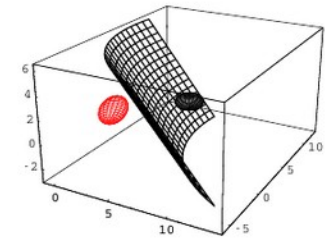
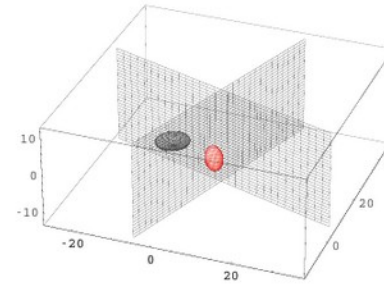
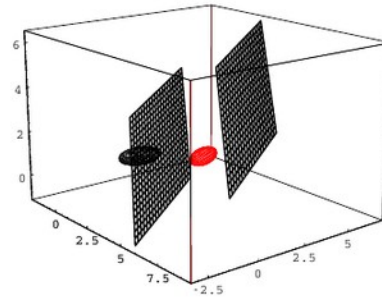
Case 3: $\boldsymbol{\Sigma}_i$ arbitrary



Discriminant Functions

$$g_i(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i) - \frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln |\boldsymbol{\Sigma}_i| + \ln(P(\omega_i))$$

Case 3: $\boldsymbol{\Sigma}_i$ arbitrary



Summary

- Elementary idea: **minimize the risk** by choosing the action which has the smallest conditional risk $R(\alpha|x)$.
- For minimization of the error probability choose the state of nature which **maximizes the a posteriori** probability $P(\omega_j|x)$.
- The main problem often is that the **conditional probabilities** $p(x|\omega_j)$ are **not known**.
- The **shape** is sometimes known, but **not the parameters**.
- In a classical case it is known that a multivariate **normal distribution** is given, but the **mean** value vector and the **covariance** matrix are **not** known.
- **In many cases considerably less is known.**
Then a method is necessary which is as insensitive as possible regarding the assumptions about the probability density.

Probability Estimation

Bayes' Theorem:
$$P(\omega|\mathbf{x}) = \frac{p(\mathbf{x}|\omega) P(\omega)}{p(\mathbf{x})}$$

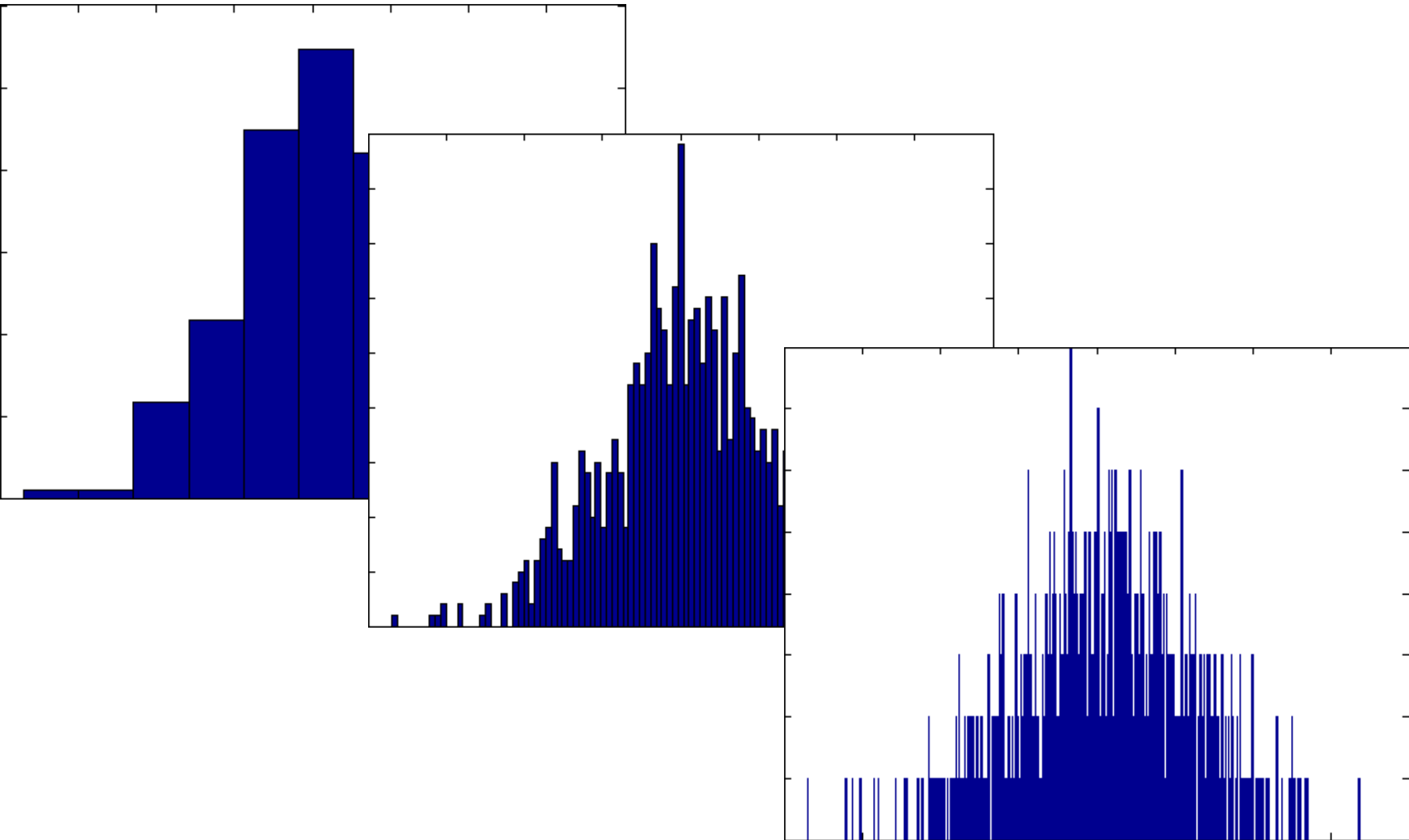
- (Bayes-)optimal classifier possible if
 - prior probability $P(\omega)$ is known → Usually given or uniform
 - class likelihood $p(\mathbf{x}|\omega)$ is known → Estimated from training data
- Training data: Measurements (images, features, etc.)
and (!) target variable (e.g. class label) are known
- Likelihood: How likely is a class given certain feature values?
How probable is a feature value given the class?
How often occurs a feature value given the class?
- Two ways of estimation: Non-parametric and parametric

Non-parametric Estimation

Histogram:

- Discrete approximation of a continuous function
- Frequentistic interpretation of probability
 - Probability is defined as ratio of number of positive events over number of possible events
- Consistent estimate
 - if $|D| \rightarrow \infty$, then $h(v)$ is true probability distribution
- Problems
 - Curse of dimensionality
 - In general: d dimensions with n bins $\rightarrow n^d$ cells
 - Number of necessary samples to estimate all parameters sufficiently well grows exponentially with the number of dimensions.
 - For non-categorical variables
 - How many bins are sufficient?
 - The more, the finer the resolution but the more data is needed

Non-parametric Estimation



Non-parametric Estimation

Parzen Windows

- Total number of samples: n
- Volume of a local (this) region: $V_n = h_n^d$
- Window function:
$$\phi(u) = \begin{cases} 1 & |u_j| \leq 0.5 \ (j=1, \dots, d) \\ 0 & \text{otherwise} \end{cases}$$
- Number of samples in a region:
$$k_n = \sum_{i=1}^n \phi\left(\frac{\mathbf{x} - \mathbf{x}_i}{h_n}\right)$$

$$p_n(\mathbf{x}) = \frac{k_n/n}{V_n} = \frac{1}{n} \sum_{i=1}^n \frac{1}{h_n^d} \phi\left(\frac{\mathbf{x} - \mathbf{x}_i}{h_n}\right)$$

Non-parametric Estimation

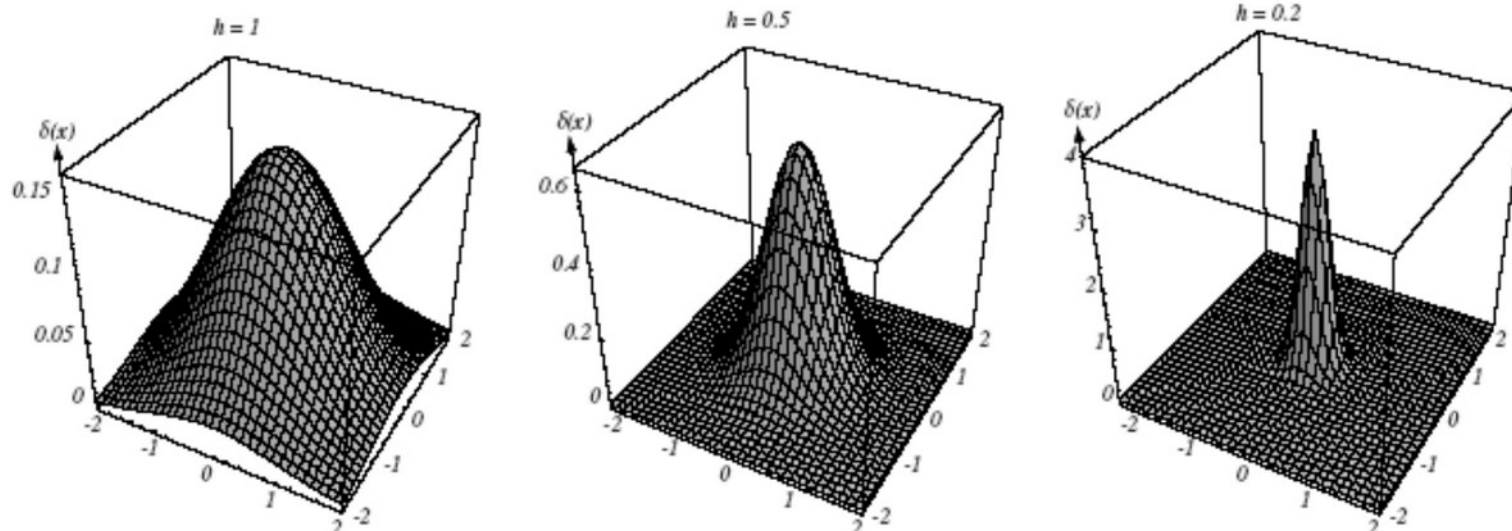
Parzen Windows

- If window function ϕ is a density function, i.e.

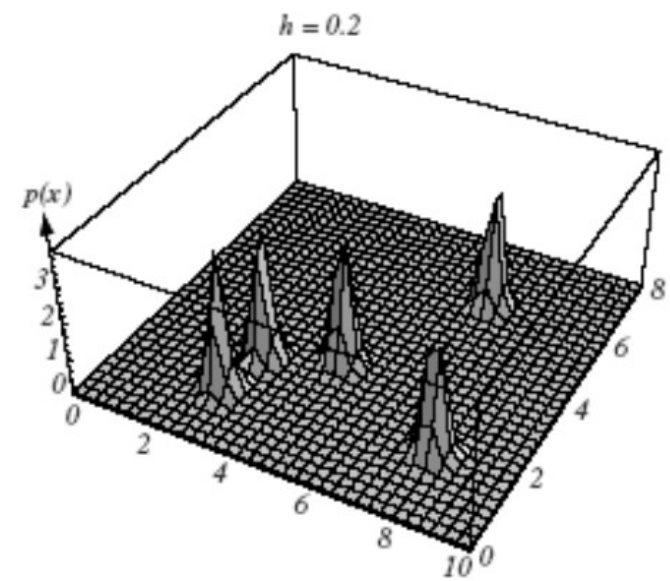
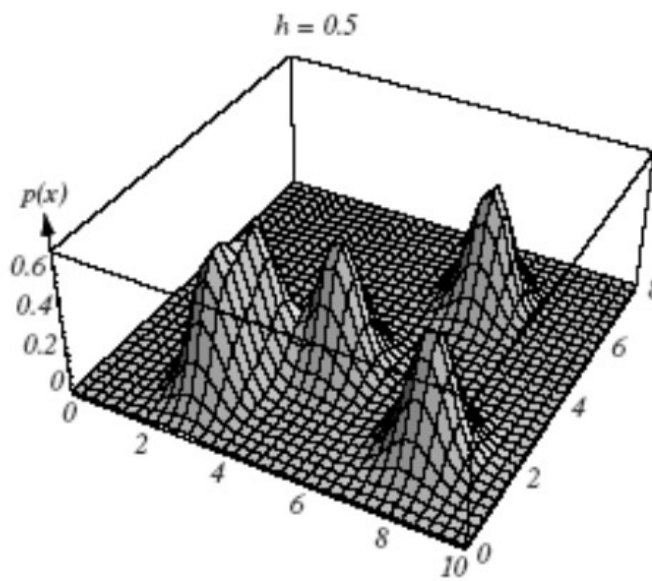
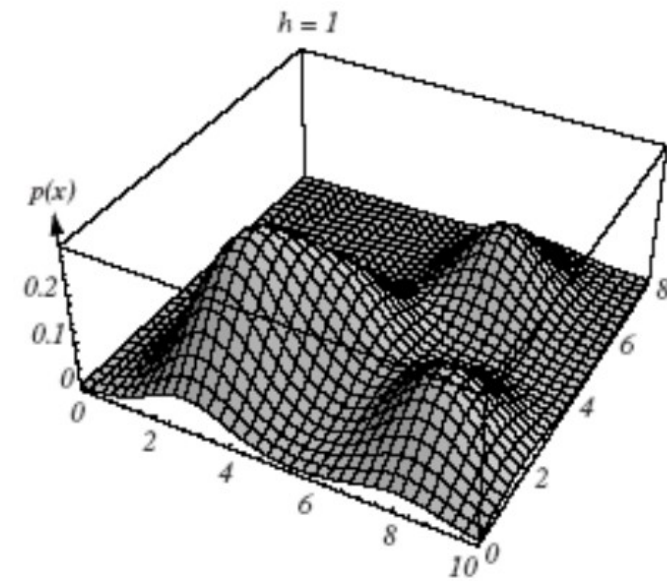
$$\phi(x) \geq 0 \quad , \quad \int \phi(u) du = 1$$

then $p_n(\mathbf{x})$ is also a density function

- Variant: k-Nearest-Neighbor estimate:
 - Fix number of samples k and vary the volume, i.e. enlarge the window until k points are contained
- Other window functions possible, e.g. Gaussian window

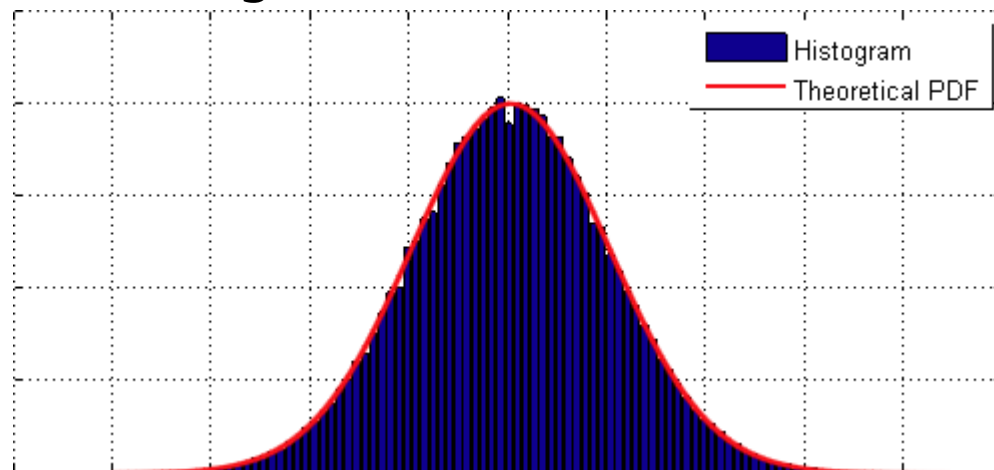


Non-parametric Estimation



Parametric Estimation

- Non-parametric density estimation approaches
 - are very general
 - make no assumptions about the actual distribution
 - need a lot of data
- Sometimes the general form of the probability density is known (or can be sufficiently approximated) and can be parameterized
 - e.g. as Gaussian function
 - Problem of estimating a huge number of “parameters” (e.g. the probability density at each point) is reformulated into estimating a few parameters (e.g. mean and variance of Gaussian function)



Maximum-Likelihood Estimation

- Assumptions, e.g. $p(\mathbf{x}|\omega_j) = p(\mathbf{x}|\omega_j, \boldsymbol{\theta}_j) \sim N(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$
 - Density can be described by a parametric model
- Task: Use training samples to estimate unknown parameter vectors
- Probability of the training data $D = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ given the model θ is:

$$p(D|\theta) = \prod_{k=1}^n p(\mathbf{x}_k|\theta)$$

→ Likelihood of the model

→ ML estimate is $\hat{\theta} = \underset{\theta}{\operatorname{argmax}} p(D|\theta)$

- Maximizes the probability of the data given the model
- Selects model that best explains data

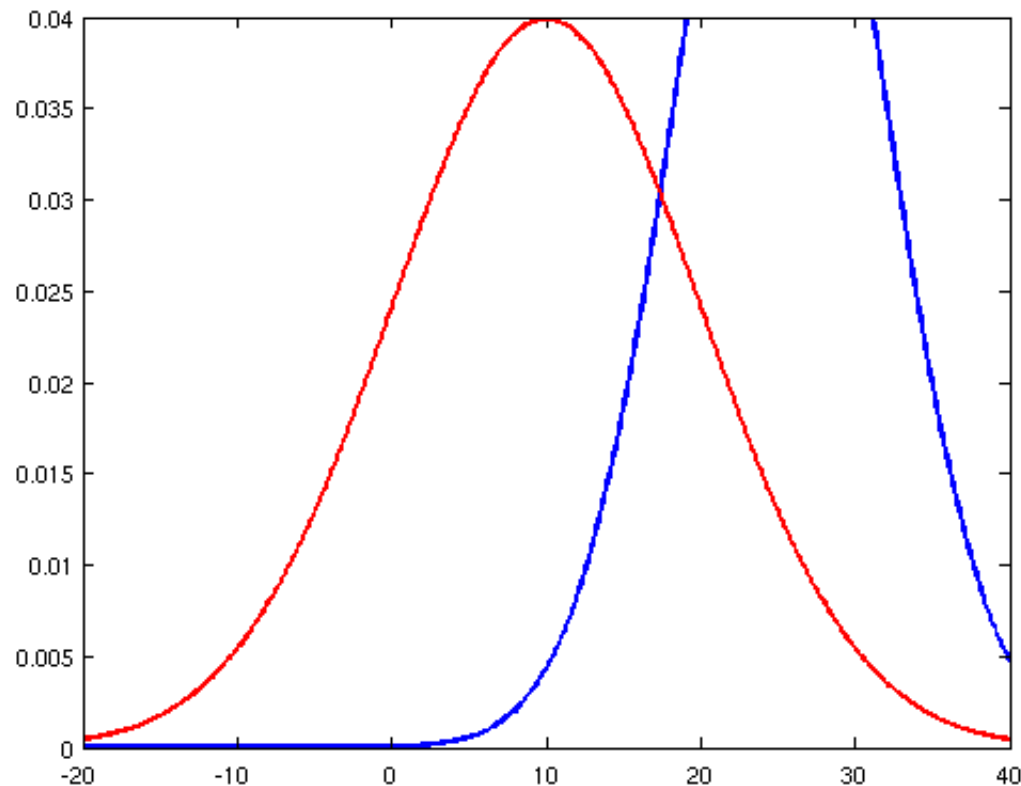
Maximum-Likelihood Estimation

- Normal distribution, i.e. $p(\mathbf{x}) = p(\mathbf{x}|\boldsymbol{\theta}) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$$

$$\hat{\mathbf{C}} = \frac{1}{n-1} \sum_{k=1}^n (\mathbf{x}_k - \hat{\boldsymbol{\mu}})(\mathbf{x}_k - \hat{\boldsymbol{\mu}})^T$$

$n = 2$



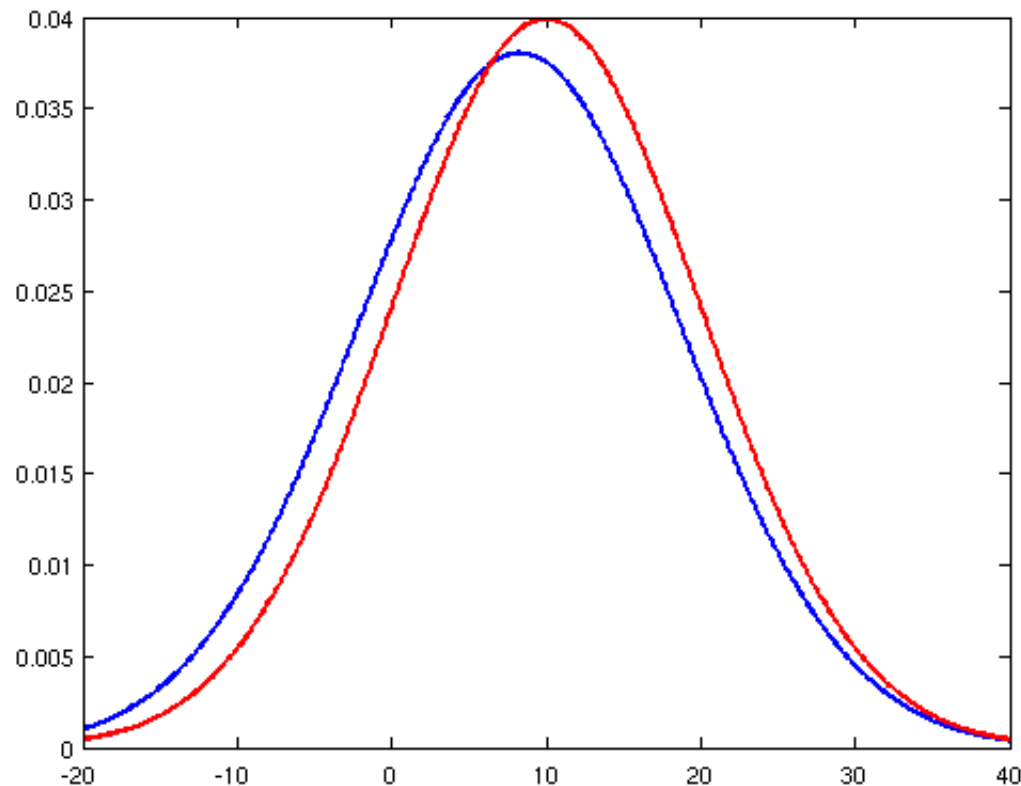
Maximum-Likelihood Estimation

- Normal distribution, i.e. $p(\mathbf{x}) = p(\mathbf{x}|\boldsymbol{\theta}) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$$

$$\hat{\mathbf{C}} = \frac{1}{n-1} \sum_{k=1}^n (\mathbf{x}_k - \hat{\boldsymbol{\mu}})(\mathbf{x}_k - \hat{\boldsymbol{\mu}})^T$$

$n = 10$



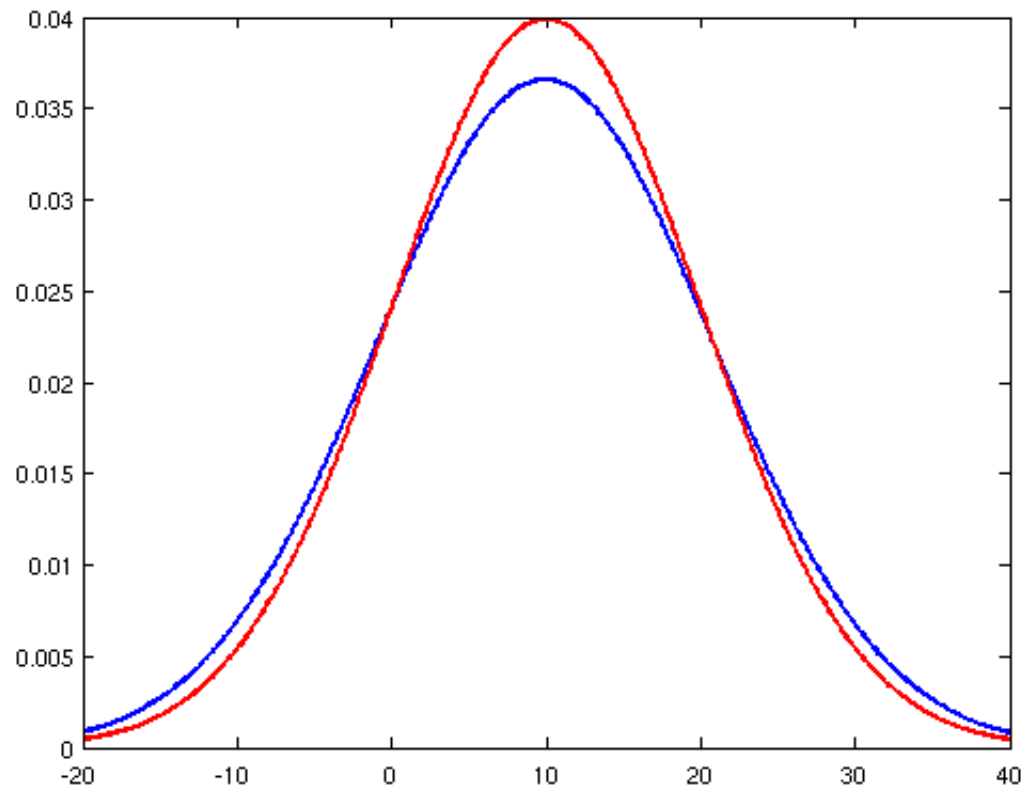
Maximum-Likelihood Estimation

- Normal distribution, i.e. $p(\mathbf{x}) = p(\mathbf{x}|\boldsymbol{\theta}) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$$

$$\hat{\mathbf{C}} = \frac{1}{n-1} \sum_{k=1}^n (\mathbf{x}_k - \hat{\boldsymbol{\mu}})(\mathbf{x}_k - \hat{\boldsymbol{\mu}})^T$$

$n = 100$



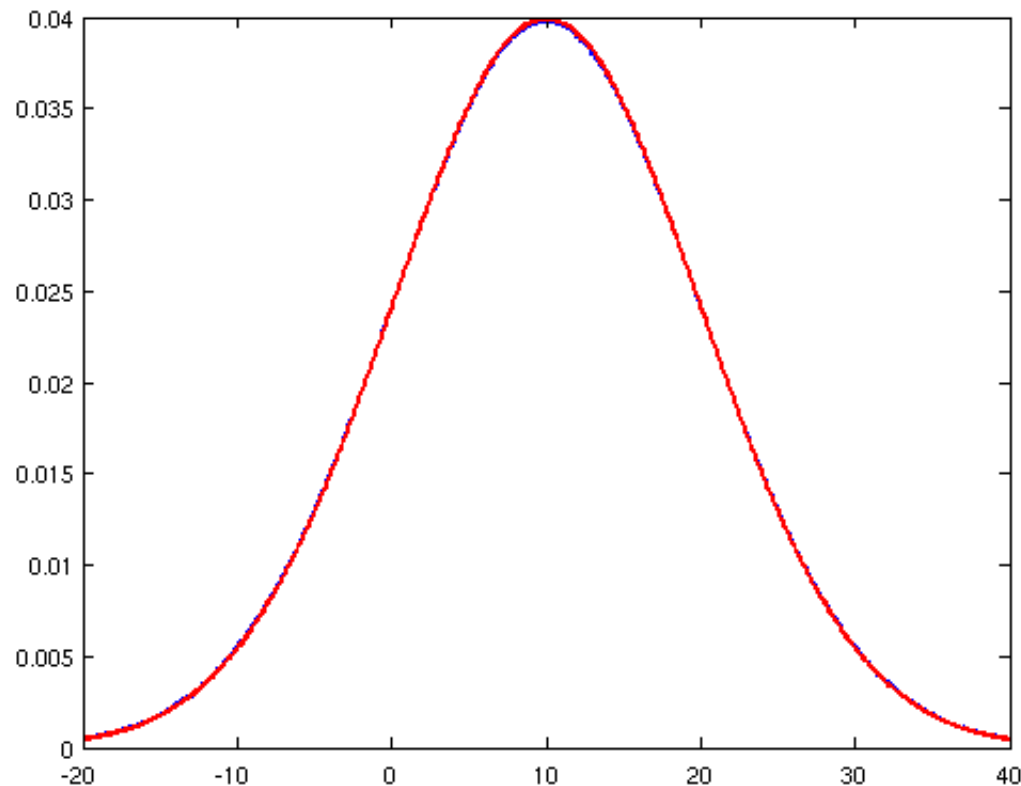
Maximum-Likelihood Estimation

- Normal distribution, i.e. $p(\mathbf{x}) = p(\mathbf{x}|\boldsymbol{\theta}) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$\hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$$

$$\hat{\mathbf{C}} = \frac{1}{n-1} \sum_{k=1}^n (\mathbf{x}_k - \hat{\boldsymbol{\mu}})(\mathbf{x}_k - \hat{\boldsymbol{\mu}})^T$$

$n = 1000$



Maximum-Likelihood Estimation

- Normal distribution, i.e. $p(\mathbf{x}) = p(\mathbf{x}|\boldsymbol{\theta}) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

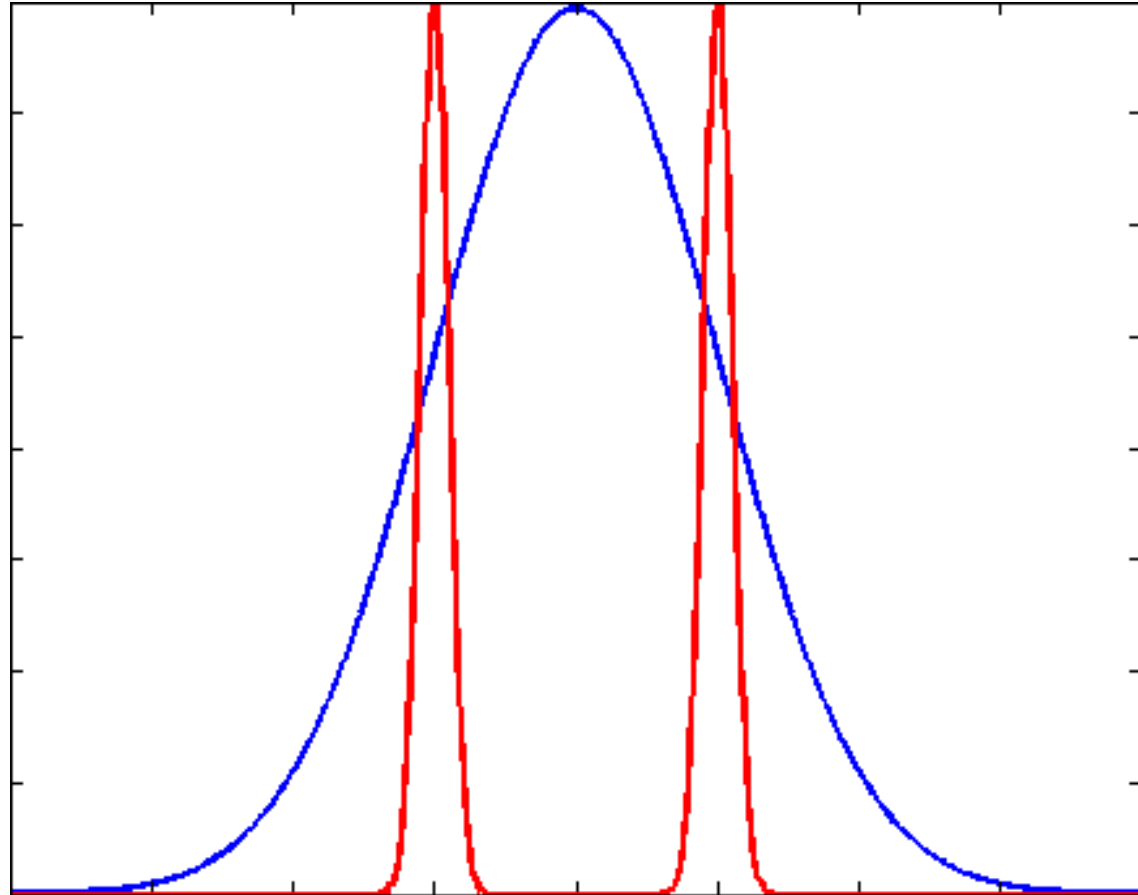
$$\hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$$

$$\hat{\mathbf{C}} = \frac{1}{n-1} \sum_{k=1}^n (\mathbf{x}_k - \hat{\boldsymbol{\mu}})(\mathbf{x}_k - \hat{\boldsymbol{\mu}})^T$$

Another example:

$n = 1,000,000$

Good or bad?



Bayesian Estimation

- Goal: Compute $P(\omega_i | \mathbf{x}) \propto p(\mathbf{x} | \omega_i) P(\omega_i)$
 - But: Likelihood and prior are unknown
 - Take prior knowledge into account
 - Function describing the shape of unknown densities
 - Value ranges of known parameters
 - Training samples

$$\rightarrow P(\omega_i | \mathbf{x}, D) = \frac{p(\mathbf{x} | \omega_i, D) P(\omega_i)}{p(\mathbf{x} | D)} = \frac{p(\mathbf{x} | \omega_i, D) P(\omega_i)}{\sum_i p(\mathbf{x} | \omega_i, D) P(\omega_i)}$$

→ This is unknown,
but assumed to follow a known parametric shape $p(\mathbf{x} | \boldsymbol{\theta}_i)$

- Known information about θ in a priori density $p(\boldsymbol{\theta}_i)$
- There is hope that the posteriori density $p(\boldsymbol{\theta}_i | D)$ has a sharp maximum at the point of the true value of $\boldsymbol{\theta}_i$.

Summary

- Instead of the probability function, its parameters are searched for.
- The maximum-likelihood procedure seeks the parameter vector which best represents the training data.
- In Bayesian estimation, parameters are considered random variable. The *a posteriori* density is calculated with help of the *a priori* density and the training data.

Now: Prior Knowledge is Used to Overcome Noisy Image Data

- Prior:
Object parameters follow a
Markov Random Field model

Markov Random Field (MRF)

- **Field:** spatial (2D/3D) arrangement of variables
- **Random:** every variable takes a value randomly
- In a general random field variables are dependent on (the) other variables.
- In a **Markov** random field variables only depend on neighboring variables.

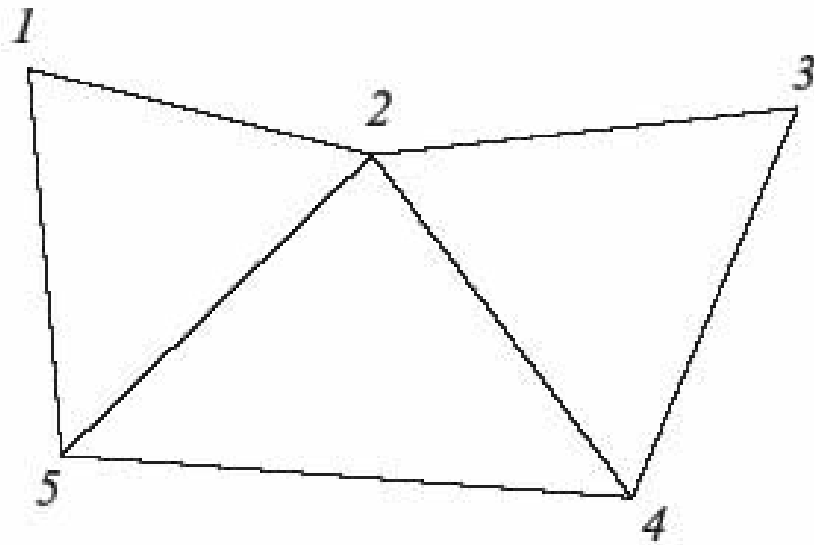
Analogy between Real World and MRF

In the real world dependencies between neighboring objects do not occur between objects distant from each other.

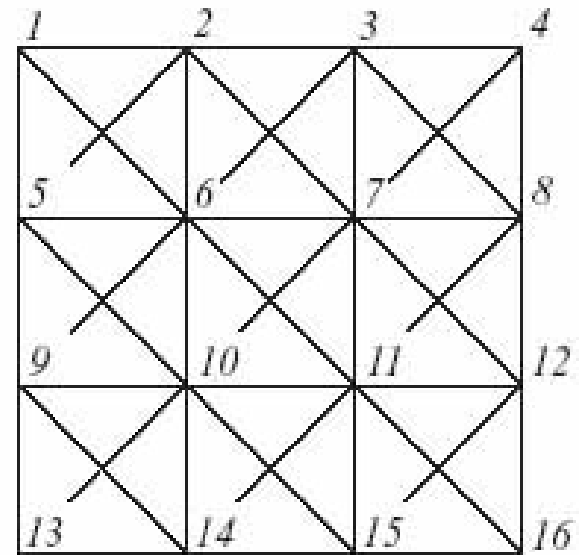
Examples:

- Buildings parallel to a street
- Locations (pixel) of a certain land use constitute larger coherent areas.
- Parts of a road are connected and constitute a long road or a road network.

Markov Random Fields and Neighborhood Gibbs Fields II



a) graph-based



b) raster-based

Markov Random Fields and Neighborhood Gibbs Fields III

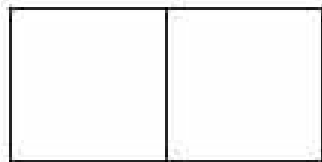
- **Neighborhood relation:** shown as a line connecting two primitives
- **Neighborhood:** all primitives to which a primitive has a neighborhood relation
- **Neighborhood system:** set of all neighborhoods of the random field
- **Clique:** Subset of a neighborhood containing only neighboring primitives
- **Potential:** Energy as a function of the states of the primitives of a clique.

Example: MRF Model for Horizontal Lines I

- Horizontal one-pixel-wide lines appear preferably in the (example) world.
- Line interruptions and line ends appear rarely.
- Selection of a Markov random field model
 - Pixels of the image as primitives of a raster-based MRF
 - Four-neighborhood system
 - Horizontal and vertical two-pixel cliques
 - Potentials of the two-pixel-cliques such that line interruptions and line ends appear rarely

Example: MRF Model for Horizontal Lines II

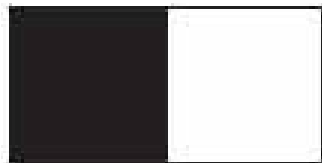
Potentials of the two-pixel cliques



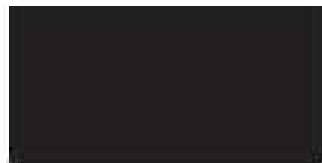
$$H_c = 0$$



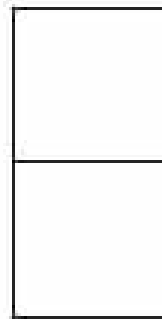
$$H_c = \alpha$$



$$H_c = \alpha$$



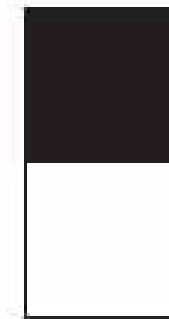
$$H_c = 0$$



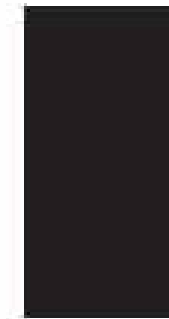
$$H_c = 0$$



$$H_c = 0$$



$$H_c = 0$$



$$H_c = \beta$$

with e.g. $\alpha = \beta = 1$

From Potential H_c to Probability Density $p(\mathbf{x})$

- Energy

$$H(\mathbf{x}) = \sum_{c \in C} H_c(\mathbf{x})$$

C : set of all cliques

- Joint probability density

$$p(\mathbf{x}) = \frac{1}{Z} \exp \left\{ -\frac{H(\mathbf{x})}{T} \right\}$$

T : temperature parameter
for simulated annealing

- Normalization constant

$$Z = \sum_{\mathbf{x} \in \Xi} \exp \left\{ -\frac{H(\mathbf{x})}{T} \right\}$$

Ξ : configuration space

From Potential H_c to Probability Density $p(x_s | \partial x_s)$

- Local energy $H_s(x_s | \partial x_s) = \sum_{c \in C_s} H_{c_s}(x_s | \partial x_s)$

C_s : set of all cliques containing primitive s

- Local conditional probability density

$$p(x_s | \partial x_s) = \frac{1}{Z_s} \exp \left\{ \frac{-H_s(x_s | \partial x_s)}{T} \right\}$$

T : temperature parameter for simulated annealing

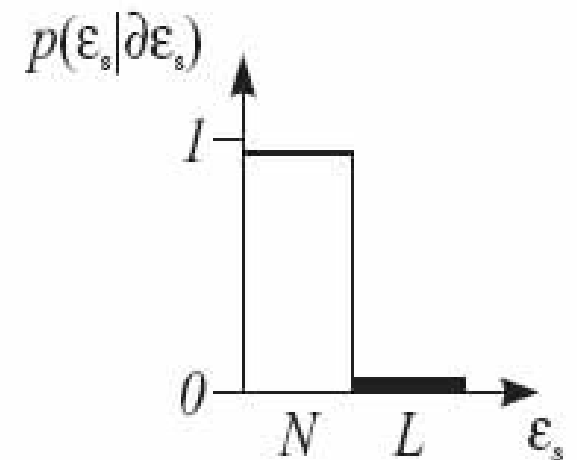
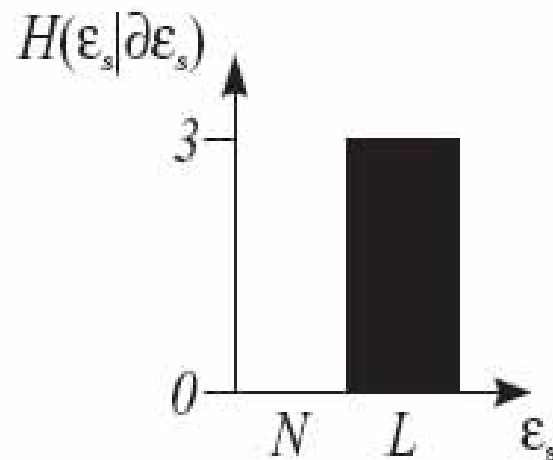
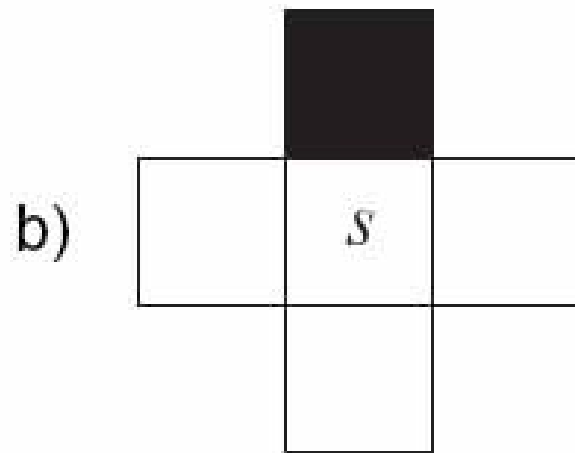
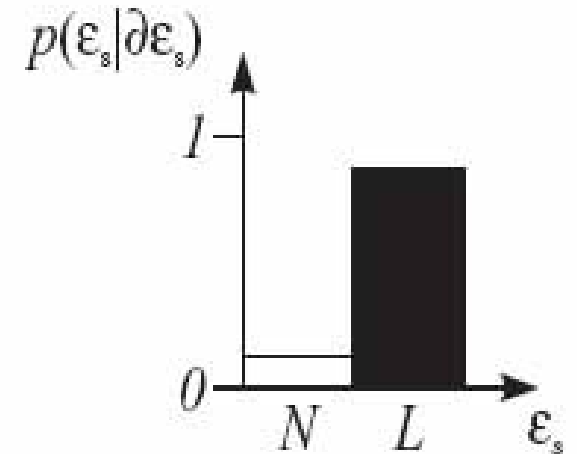
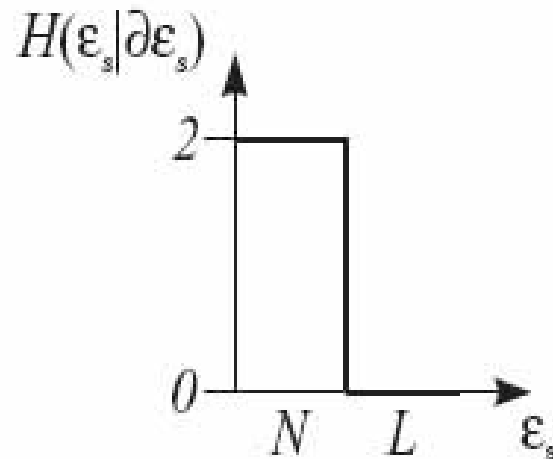
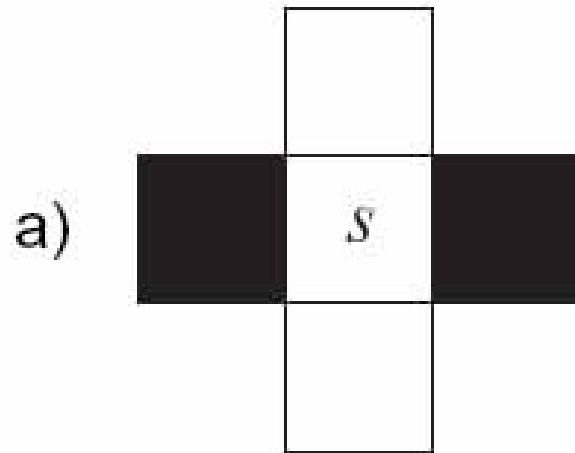
- Normalization constant

$$Z_s = \sum_{\xi \in \Xi_s} \exp \left\{ \frac{-H_s(\xi | \partial x_s)}{T} \right\}$$

Ξ_s : state space

Example: MRF Model for Horizontal Lines III

Determination of the local conditional probability density



Estimation of Object Parameters

- Repetition: Bayesian theorem

$$p(\boldsymbol{\varepsilon}|\boldsymbol{y}) \propto p(\boldsymbol{y}|\boldsymbol{\varepsilon}) p(\boldsymbol{\varepsilon})$$

- Bayesian theorem for local characteristics of the Markov random field

$$p(\varepsilon_s | \partial \varepsilon_s, \boldsymbol{y}_s) \propto p(\boldsymbol{y}_s | \varepsilon_s) \cdot p(\varepsilon_s | \partial \varepsilon_s)$$

- Estimation methods (examples):
 - Maximum Likelihood (ML) estimation
 - Maximum A Posteriori (MAP) estimation

Example: MRF Model for Horizontal Lines VIII

Estimation of object parameters

- Likelihood function

$$p(y_s | \varepsilon_s) = \frac{1}{\sqrt{2\pi\sigma_\varepsilon}} \exp \left\{ -\frac{(y_s - \mu_\varepsilon)^2}{2\sigma_\varepsilon^2} \right\}$$

- Model parameters (likelihood function):

$$\mu_L = 90, \sigma_L = 10, \mu_N = 110, \sigma_N = 10$$

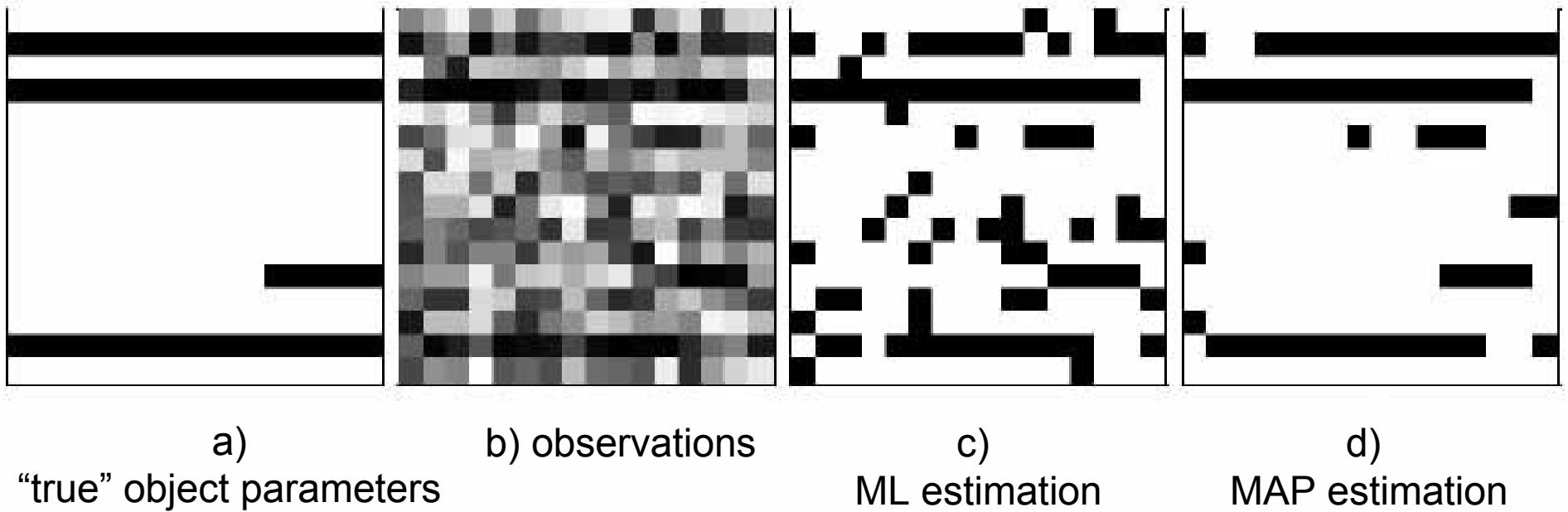
- Model parameters (a priori density): $\alpha = 1, \beta = 1$

- Temperature:

$$T = \frac{0.7}{\ln(i)}$$

Example: MRF Model for Horizontal Lines IX

Estimation of object parameters

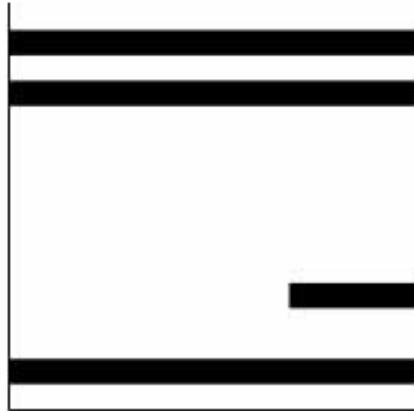


Estimation of Model Parameters

- E.g. for adaptation of the model to individual datasets
- Algorithm which uses image data and “true” object parameters as training data
- Simulated annealing can be used to estimate optimal model parameters.
- Evaluation of the difference between estimation results and “true” parameters
- Evaluation results of successive iterations control simulated annealing.

Example: MRF Model for Horizontal Lines X

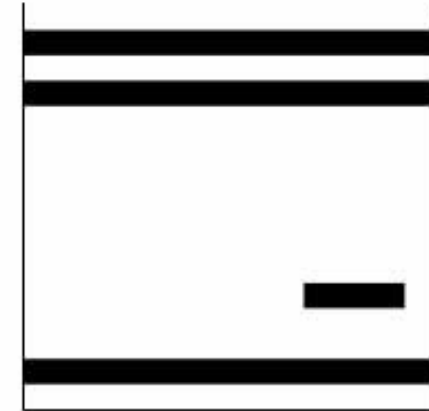
Estimation of model parameters



"true" object parameters



MAP estimation
without model parameter
optimization:
 $\alpha = 1.0, \beta = 1.0$



MAP estimation
with model parameter optimization:
 $\alpha = 2.2, \beta = 1.0$

Further Reading

- Duda, R. O.; Hart, P. E.; Stork, D.G.: Pattern Classification. 2nd edition, John Wiley & Sons, New York 2001.
- Li, S. Z. (2012). Markov random field modeling in computer vision. Springer Science & Business Media.
- Zheng, S., Jayasumana, S., Romera-Paredes, B., Vineet, V., Su, Z., Du, D., ... & Torr, P. H. (2015). Conditional random fields as recurrent neural networks. In Proceedings of the IEEE international conference on computer vision (pp. 1529-1537).
- Gawlikowski, J., Tassi, C. R. N., Ali, M., Lee, J., Humt, M., Feng, J., ... & Zhu, X. X. (2023). A survey of uncertainty in deep neural networks. Artificial Intelligence Review, 56(Suppl 1), 1513-1589.